

TROLL STATION, Antarctica

WMO: 895040

Lat: 72.012S

Lon: 2.541E

Elev: 4190

StdP: 12.60

Time Zone: 0.00 (TRL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -35.4	(c) -32.6	(d) -48.8	(e) 0.4	(f) -32.0	(g) -46.2	(h) 0.4	(i) -28.9	(j) 53.3	(k) 5.2	(l) 47.3	(m) 1.5	(n) 3.2	(o) 230	(p) 1.265

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 10.2	(c) 32.4	(d) 24.0	(e) 29.2	(f) 22.0	(g) 27.0	(h) 20.7	(i) 22.9	(j) 28.5	(k) 21.5	(l) 27.0	(m) 20.2	(n) 25.4	(o) 8.7	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	14.0	22.9	13.3	12.6	22.2	11.4	11.5	21.4	8.7	28.8	8.2	27.3	7.7	25.6	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 50.7	(b) 41.7	(c) 33.1	(e) -39.2	(f) 34.3	(g) 3.8	(h) 2.2	(i) -41.9	(j) 35.9	(k) -44.2	(l) 37.2	(m) -46.3	(n) 38.5	(o) -49.1	(p) 40.1		
			DB		-39.3		25.7		4.0		1.5					
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	19.7	14.5	3.9	-3.6	-5.9	-9.4	-13.5	-12.4	-11.3	-1.4	10.3	19.6
	DBStd	14.57	4.29	6.09	6.90	9.50	10.34	9.02	10.48	10.85	10.20	9.12	5.75	4.08
	HDD50	17952	939	993	1429	1608	1733	1782	1967	1936	1838	1594	1190	943
	HDD65	23427	1404	1413	1894	2058	2198	2232	2432	2401	2288	2059	1640	1408
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.8	9.1	9.2	7.1	8.4	9.5	7.8	7.4	9.0	7.8	9.5	10.6	10.2
Precipitation	PrecAvg	10.7	0.7	0.8	0.7	0.7	1.1	0.9	0.8	1.0	0.7	1.0	1.3	0.9
	PrecMax	13.6	1.7	1.7	1.7	1.4	2.6	2.3	1.7	2.7	1.7	2.1	3.1	2.3
	PrecMin	7.7	0.2	0.2	0.2	0.2	0.4	0.0	0.2	0.3	0.1	0.4	0.0	0.2
	PrecStd	1.7	0.4	0.4	0.4	0.4	0.5	0.5	0.4	0.6	0.4	0.4	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	39.3	24.7	21.0	20.5	14.8	14.5	17.7	15.4	21.3	28.4	35.5
		MCWB	26.6	27.7	18.8	17.9	15.9	11.6	11.2	13.7	10.7	15.3	20.0	25.0
	2%	DB	32.3	31.7	19.4	15.8	15.8	10.6	9.7	12.5	11.9	18.0	24.3	30.8
		MCWB	24.1	23.3	14.6	11.7	12.1	8.1	7.2	8.7	8.2	12.9	17.9	23.1
	5%	DB	29.1	27.4	16.7	13.1	12.3	7.1	5.5	8.6	7.7	14.5	21.5	28.2
		MCWB	22.3	20.4	12.6	9.2	9.3	4.5	3.1	5.6	4.9	10.3	15.8	21.2
	10%	DB	26.9	23.8	14.3	10.6	9.3	4.1	2.1	3.9	4.5	11.7	19.2	26.2
		MCWB	21.1	18.2	10.5	7.4	6.6	1.8	-0.1	1.3	1.8	8.4	14.3	20.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	22.6	17.3	19.0	16.7	12.0	11.6	14.4	10.9	15.9	20.1	24.7
		MCDB	32.3	30.3	22.2	21.2	19.9	14.1	14.1	17.7	14.6	20.7	26.6	30.4
	2%	WB	23.2	20.2	14.4	12.2	12.5	8.3	7.2	8.9	8.0	12.9	17.6	22.6
		MCDB	28.9	26.0	18.2	16.1	15.1	10.5	9.7	12.0	11.5	17.5	23.3	28.0
	5%	WB	22.1	18.4	12.4	9.6	9.4	4.6	3.2	5.6	5.1	10.6	15.6	21.1
		MCDB	27.4	23.2	15.9	13.1	12.0	7.0	5.6	8.7	7.8	14.3	20.2	26.7
	10%	WB	20.8	16.9	10.1	7.6	6.8	1.9	0.0	1.3	1.9	8.3	13.9	19.9
		MCDB	25.8	21.2	13.4	10.8	9.2	4.1	2.1	3.9	4.6	11.6	18.2	25.1
Mean Daily Temperature Range	5% DB	MDBR	10.2	12.4	12.4	12.0	12.5	12.4	12.6	13.1	13.7	13.3	11.1	9.4
		MCDBR	14.9	19.9	14.1	12.0	11.4	13.7	12.0	14.0	14.0	13.5	15.1	12.7
		MCWBR	6.8	8.7	8.6	9.5	9.5	12.0	10.6	12.2	12.0	10.8	9.8	6.8
	5% WB	MCDBR	9.9	10.9	10.6	11.6	10.8	13.0	11.8	13.8	13.0	13.0	12.2	9.8
		MCWBR	6.7	7.8	8.6	9.7	9.2	11.5	10.6	12.1	11.1	10.5	9.2	6.8
Clear-Sky Solar Irradiance	taub	0.199	0.187	0.165	0.109	N/A	N/A	N/A	N/A	0.106	0.151	0.169	0.187	0.199
	taud	2.292	2.302	2.271	1.973	N/A	N/A	N/A	N/A	1.942	2.208	2.251	2.271	2.276
	Ebn at Noon	335	320	291	211	0	0	0	0	206	293	326	339	340
	Edh at Noon	33	27	19	8	0	0	0	0	8	19	28	33	35
All-Sky Solar Radiation	RadAvg	2810	1820	841	181	0	0	0	0	68	545	1436	2511	3132
	RadStd	58	58	33	8	0	0	0	0	4	25	46	88	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page